La Scuola d'Italia / Dr. Huson / IB Math 11
2 November 2026

Nickname:

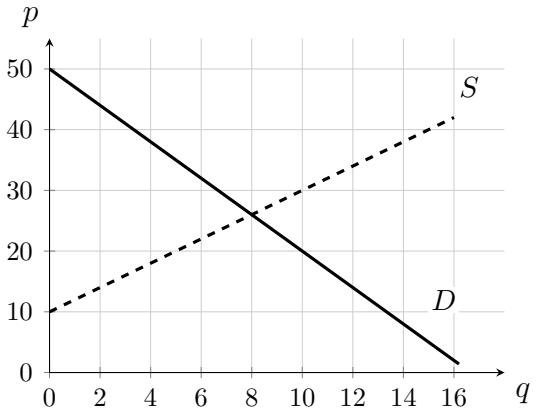
2.7 Challenge: Systems of Equations and Market Equilibrium (no calculator)

1. The market for phone cases in a city is modelled by two straight lines. The quantity is q thousand cases per month and the price is p euros per case.

   Demand line D: $p = 50 - 3q$

   Supply line S: $p = 10 + 2q$

   The market is in *equilibrium* at the point where the two lines intersect.



   (a) Use the graph to estimate the equilibrium quantity and price.

   (b) Write both equations in the form $aq + bp = c$. Hence, use the elimination method to find the exact equilibrium quantity and price.

   (c) Write down the gradient of the supply line. Interpret this value in context.

   (d) The city sets a maximum price of 20 euros per case. Find the quantity demanded and the quantity supplied at this price. Hence, find the shortage of phone cases.

   (e) The *consumer surplus* is the area of the triangle between the demand line, the p-axis and the horizontal line at the equilibrium price. Shade this triangle on the graph and find the consumer surplus in euros.

2. Giulia invests 10 000 euros for one year. She puts x euros into a savings bond that pays 3% interest per year and the remaining y euros into a fund that pays 7% per year. After one year she has earned 540 euros in interest.

   (a) Write down two equations in x and y.

   (b) Use the elimination method to find the amount invested in the bond and the amount invested in the fund.

   Extension. Marco invests 20 000 euros for one year in three funds. He puts x euros into a fund paying 2% per year, y euros into a fund paying 4% and z euros into a fund paying 6%. He earns 740 euros in interest. He puts 2000 euros more into the 4% fund than into the 6% fund.

   (c) (i) Write down three equations in x, y and z.

   (ii) Use the elimination method to find x, y and z.